



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
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Course Code: CSA0509

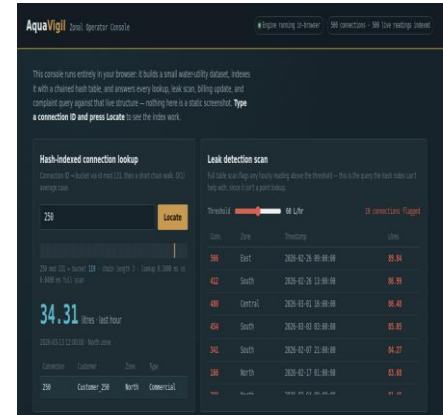
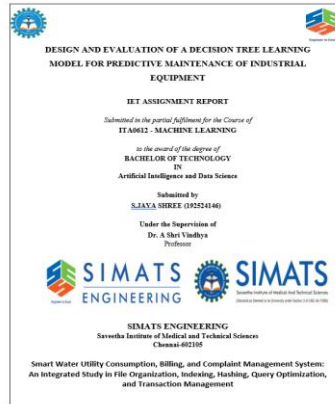
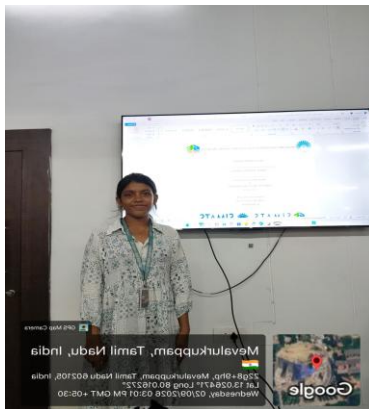
Slot: A

Course Name: Database Management systems

Course Faculty: Dr.R.Kesavan

Project Title: Smart Water Utility Consumption, Billing, and Complaint Management System: An Integrated Study in File Organization, Indexing, Hashing, Query Optimization, and Transaction Management

Module Photographs: (3 photographs –Module Photo, Individual student contribution module work in the project and presentation image)



Project Description: (here you write what you did in this project (contribution) including Model Description)

Our solution is a Smart Water Utility Consumption, Billing, and Complaint Management System developed to improve the efficiency and reliability of water utility services. This project presents an end-to-end database solution for a large-scale smart water utility, focusing on two major problems: slow retrieval of continuously growing meter-reading data and uncoordinated billing and payment operations that can cause duplicate bills and lost updates. The system combines database file organization, indexing, hashing, relational schema design, transaction management, and a web-based operator dashboard. An indexed-sequential approach is used for efficient ordered and range-based meter-reading queries, while a static hash table provides fast access to the latest reading for a connection. A composite B-tree index on (connection_id, reading_ts) improves meter-history searches, and a secondary index on consumption_ltr supports leak detection. The relational database contains six main tables: Zone, Connection, MeterReading, Bill, Payment, and ComplaintTicket, with primary keys, foreign keys, and a UNIQUE(connection_id, billing_cycle) constraint to prevent duplicate bills. ACID transactions and BEGIN IMMEDIATE locking are used to safely handle concurrent billing and payment operations. The system was implemented using SQLite and Python, with HTML, CSS, and JavaScript used to create the AquaVigil operator console.

Student Signature

Guide Signature